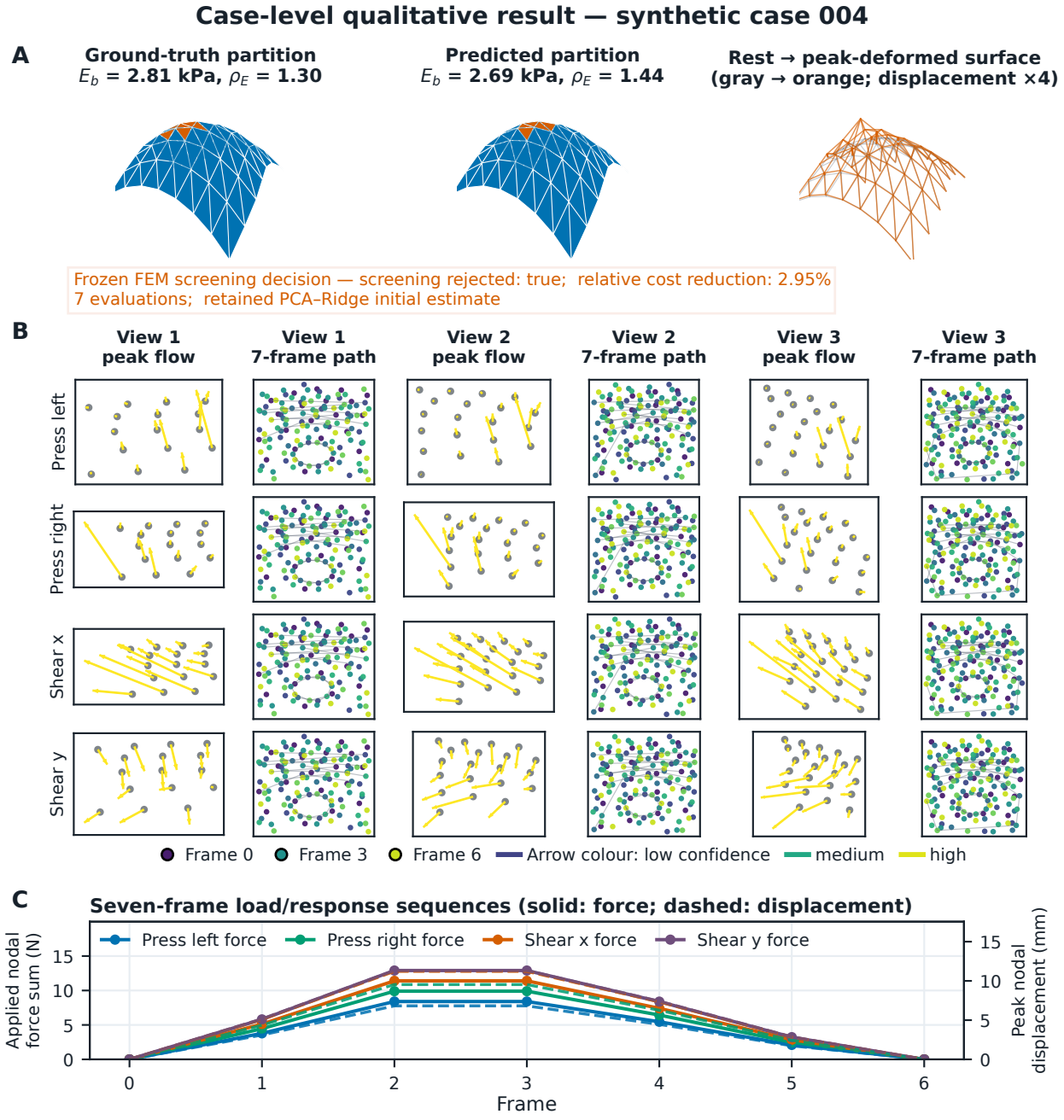

Case-level qualitative result — synthetic case 004
A
Ground-truth partition
E_b = 2.81 kPa, ρ_E = 1.30
Predicted partition
E_b = 2.69 kPa, ρ_E = 1.44
Rest → peak-deformed surface
(gray → orange; displacement ×4)
Frozen FEM screening decision — screening rejected: true; relative cost reduction: 2.95%
7 evaluations; retained PCA–Ridge initial estimate
B
View 1 peak flow
View 1 7-frame path
View 2 peak flow
View 2 7-frame path
View 3 peak flow
View 3 7-frame path
Press left
Press right
Shear x
Shear y
Frame 0
Frame 3
Frame 6
Arrow colour: low confidence
medium
high
C
Seven-frame load/response sequences (solid: force; dashed: displacement)
Press left force
Press right force
Shear x force
Shear y force
Applied nodal force sum (N)
Peak nodal displacement (mm)
Frame